

Phase 6 TSP Replay Mechanism Report

Overview

- Run count: 20.
- Condition count: 9.
- Best mean transfer gap: phase6_diversity_failure_replay.

Mechanism Answers

1. Why did random replay beat failure replay? Random replay beat raw failure replay only narrowly, but the paired comparison points in the same direction as the coverage hypothesis. Relative to no replay, random replay changed transfer by -0.003123. Relative to random replay, failure replay changed transfer by 0.001361, archive diversity by -0.065566, and archive hardness by 0.128613.
2. Does replay archive diversity explain transfer better than replay hardness? Archive hardness explains transfer at least as strongly as diversity.
3. Does compression help once applied to the best replay mechanism? Random-replay compression changed transfer by 0.014874 and complexity by -0.03 relative to plain random replay.
4. Are the final heuristics genuinely different, or just differently tuned? Final heuristics should be treated as genuinely different only when transfer, novelty, and complexity move together; the paired novelty and complexity deltas in this aggregate are the main evidence for that distinction.

Condition Comparison

Condition	Mean T ransfer Gap	Mean TSPLI B Gap	Mean S ynthetic Gap	Archive Diversit y	Archiv e Hard ness	Failure C oncentrat ion	Mean Accepted Novelty	Mean Compl exity	Adaptatio n Efficiency
phase6_diversity_failure_replay	0.089519	0.132031	0.015122	0.171206	0.279086	0.172965	0.652934	0.72	0.023136
phase6_diversity_failure_replay_compression	0.111045	0.161437	0.022859	0.171206	0.288591	0.172553	0.284606	0.7	0.017447
phase6_diversity_weighted_replay	0.1035	0.152842	0.017152	0.236772	0.180063	0.143749	0.643928	0.72	0.022408
phase6_failure_replay	0.098156	0.142676	0.020247	0.171206	0.296396	0.234171	0.639973	0.7	0.016269
phase6_no_replay	0.099918	0.14391	0.022932	0.236772	0.179815	0.0	0.74868	0.7035	0.015112
phase6_random_replay	0.096795	0.143616	0.014856	0.236772	0.167783	0.19457	0.676544	0.72	0.029828
phase6_random_replay_compression	0.111669	0.161783	0.023969	0.236772	0.181505	0.19457	0.395132	0.69	0.015059
phase6_residual_failure_replay	0.120734	0.172899	0.029446	0.118795	0.278985	0.299628	0.548243	0.68	-0.002814
phase6_stratified_random_replay	0.095646	0.141696	0.015057	0.236772	0.174478	0.221162	0.62979	0.72	0.020423

Archive-Diversity Diagnostics

- Diversity vs TSPLIB gap Pearson r: -0.009478.
- Diversity vs TSPLIB gap Spearman rho: -0.041013.
- Hardness vs TSPLIB gap Pearson r: 0.39687.
- Hardness vs TSPLIB gap Spearman rho: 0.371779.

Paired Comparisons

Negative deltas favor the candidate for transfer gap, benchmark gap, synthetic gap, hardness, concentration, novelty, and complexity. Positive deltas favor the candidate for archive diversity and adaptation efficiency.

Comparison	Paired Runs	Transfer Delta	TSPLIB Delta	Archive Diversity Delta	Archive Hardness Delta	Novelty Delta	Complexity Delta	Adaptation Delta
phase6_random_replay vs phase6_no_replay	20	-0.003123	-0.000293	0.0	-0.012032	-0.072135	0.0165	0.014716
phase6_failure_replay vs phase6_random_replay	20	0.001361	-0.000941	-0.065566	0.128613	-0.036571	-0.02	-0.013558
phase6_random_replay_compression vs phase6_random_replay	20	0.014874	0.018166	0.0	0.013722	-0.281412	-0.03	-0.014769
phase6_stratified_random_replay vs phase6_random_replay	20	-0.001149	-0.001921	0.0	0.006695	-0.046754	0.0	-0.009405
phase6_diversity_weighted_replay vs phase6_random_replay	20	0.006706	0.009226	0.0	0.01228	-0.032616	0.0	-0.007419
phase6_residual_failure_replay vs phase6_failure_replay	20	0.022578	0.030223	-0.052411	-0.017411	-0.09173	-0.02	-0.019083
phase6_diversity_failure_replay vs phase6_failure_replay	20	-0.008637	-0.010644	0.0	-0.01731	0.012961	0.02	0.006866
phase6_diversity_failure_replay_compression vs phase6_diversity_failure_replay	20	0.021526	0.029405	0.0	0.009505	-0.368327	-0.02	-0.005689

Paired Notes

- phase6_random_replay vs phase6_no_replay used 20 paired runs.
- Transfer delta -0.003123 with 95% CI -0.027964 to 0.021061.
- TSPLIB delta -0.000293 with 95% CI -0.032786 to 0.031526.
- Archive-diversity delta 0.0 with 95% CI 0.0 to 0.0.
- phase6_failure_replay vs phase6_random_replay used 20 paired runs.
- Transfer delta 0.001361 with 95% CI -0.022536 to 0.025457.
- TSPLIB delta -0.000941 with 95% CI -0.028763 to 0.027462.
- Archive-diversity delta -0.065566 with 95% CI -0.065566 to -0.065566.
- phase6_random_replay_compression vs phase6_random_replay used 20 paired runs.

- Transfer delta 0.014874 with 95% CI -0.006282 to 0.035586.
- TSPLIB delta 0.018166 with 95% CI -0.008968 to 0.044141.
- Archive-diversity delta 0.0 with 95% CI 0.0 to 0.0.
- phase6_stratified_random_replay vs phase6_random_replay used 20 paired runs.
- Transfer delta -0.001149 with 95% CI -0.023074 to 0.02094.
- TSPLIB delta -0.001921 with 95% CI -0.029404 to 0.02615.
- Archive-diversity delta 0.0 with 95% CI 0.0 to 0.0.
- phase6_diversity_weighted_replay vs phase6_random_replay used 20 paired runs.
- Transfer delta 0.006706 with 95% CI -0.020065 to 0.033445.
- TSPLIB delta 0.009226 with 95% CI -0.023988 to 0.041405.
- Archive-diversity delta 0.0 with 95% CI 0.0 to 0.0.
- phase6_residual_failure_replay vs phase6_failure_replay used 20 paired runs.
- Transfer delta 0.022578 with 95% CI -0.005643 to 0.04867.
- TSPLIB delta 0.030223 with 95% CI -0.004028 to 0.062114.
- Archive-diversity delta -0.052411 with 95% CI -0.073656 to -0.033974.
- phase6_diversity_failure_replay vs phase6_failure_replay used 20 paired runs.
- Transfer delta -0.008637 with 95% CI -0.035972 to 0.017862.
- TSPLIB delta -0.010644 with 95% CI -0.044114 to 0.021829.
- Archive-diversity delta 0.0 with 95% CI 0.0 to 0.0.
- phase6_diversity_failure_replay_compression vs phase6_diversity_failure_replay used 20 paired runs.
- Transfer delta 0.021526 with 95% CI -0.004764 to 0.047151.
- TSPLIB delta 0.029405 with 95% CI -0.003787 to 0.06157.
- Archive-diversity delta 0.0 with 95% CI 0.0 to 0.0.